THEORY

LAN and WAN · the road analogy

Use the road comparison to distinguish networks and their connections.

WHAT IS A NETWORK?

Two or more computers or devices connected to share data, resources or services. A network may use cables, wireless links, or both.

THE ROAD ANALOGY

LAN · PRIVATE ROAD OR SCHOOL CAR PARK

Picture: A small, local route between nearby places.

Technical: A Local Area Network covers a limited geographical area, such as one school site.

WAN · MOTORWAY BETWEEN CITIES

Picture: Roads link distant places and many local routes.

Technical: A Wide Area Network spans a large area and can connect several LANs.

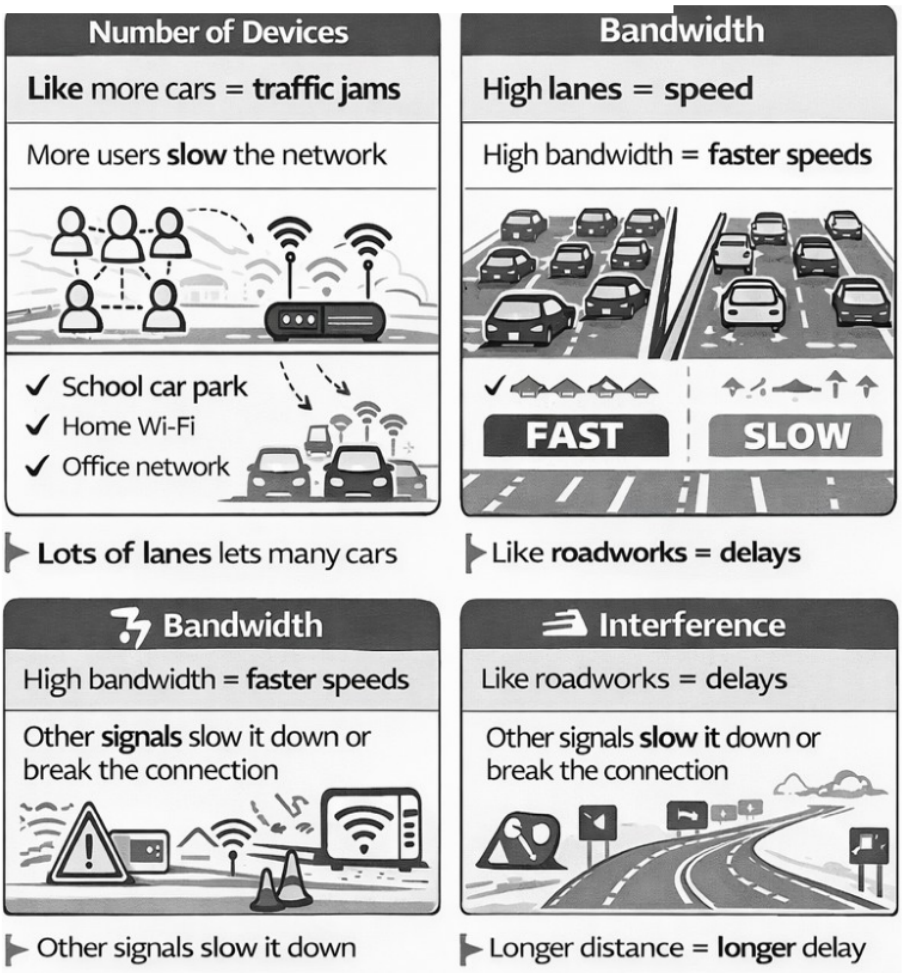
EXAM ACCURACY

LAN does not always mean “fast”, and WAN does not always mean “slow”. Actual speed depends on the connections, load, equipment and bandwidth.

THEORY

Network performance · road traffic

Each cartoon represents a different network idea. Identify what each illustrates.



BANDWIDTH = ROAD CAPACITY

Bandwidth is the maximum amount of data that can be transferred per unit of time. More bandwidth can help when multiple devices share a connection.

NUMBER OF DEVICES AND INTERFERENCE

More simultaneous traffic can cause congestion. Interference or unreliable wireless conditions may also lower actual transfer speed. Latency describes delay; it is not the same as bandwidth.

THEORY

Client–server and peer-to-peer

A school file server is a useful comparison with computers sharing files directly.

CLIENT–SERVER

Clients request services from a server. The server can provide files, accounts or web pages.

ADVANTAGE

Central accounts and backups; consistent access and management.

LIMITATION

Dedicated services cost money; some server failures can interrupt access.

PEER-TO-PEER

Each computer can directly provide resources to others and also use shared resources.

ADVANTAGE

Straightforward for a very small network; no dedicated server required.

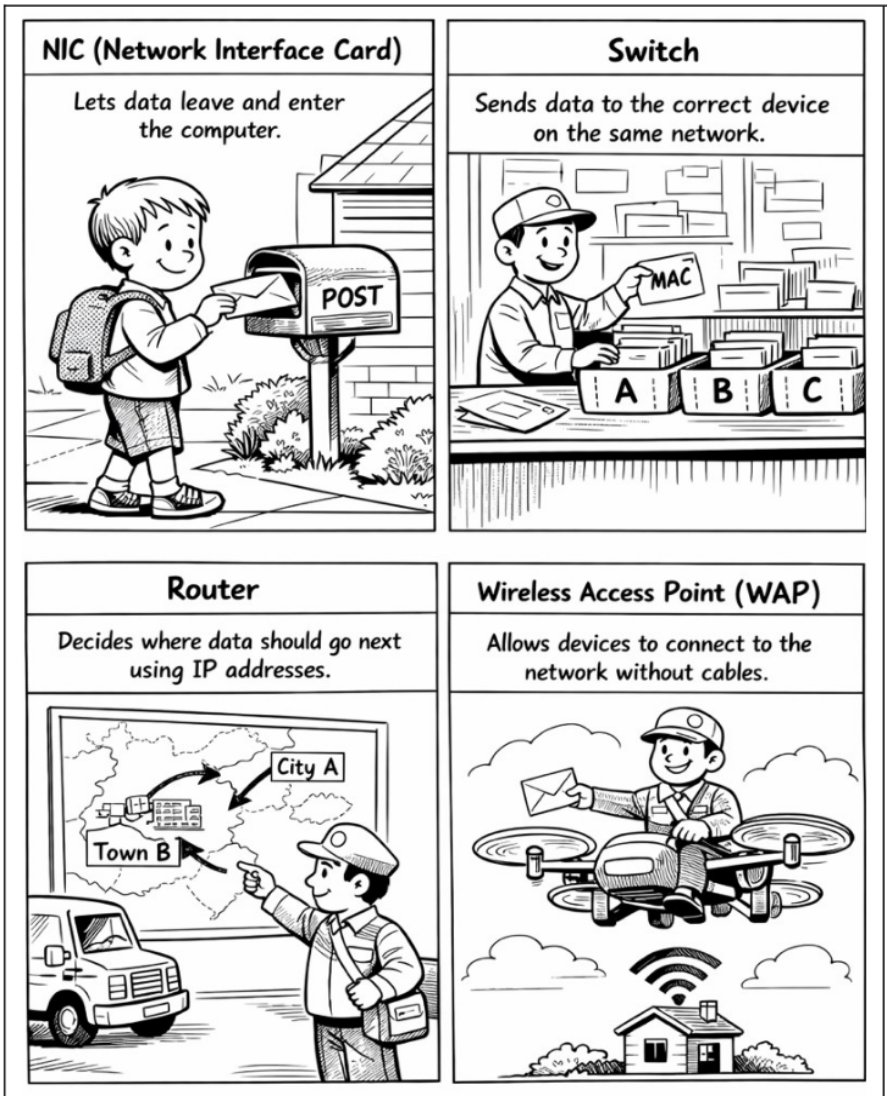
LIMITATION

No automatic central management or backup; peers must be available to share.

THEORY

Network hardware · illustrated

The postal, switch, routing and wireless delivery illustrations are kept intact.



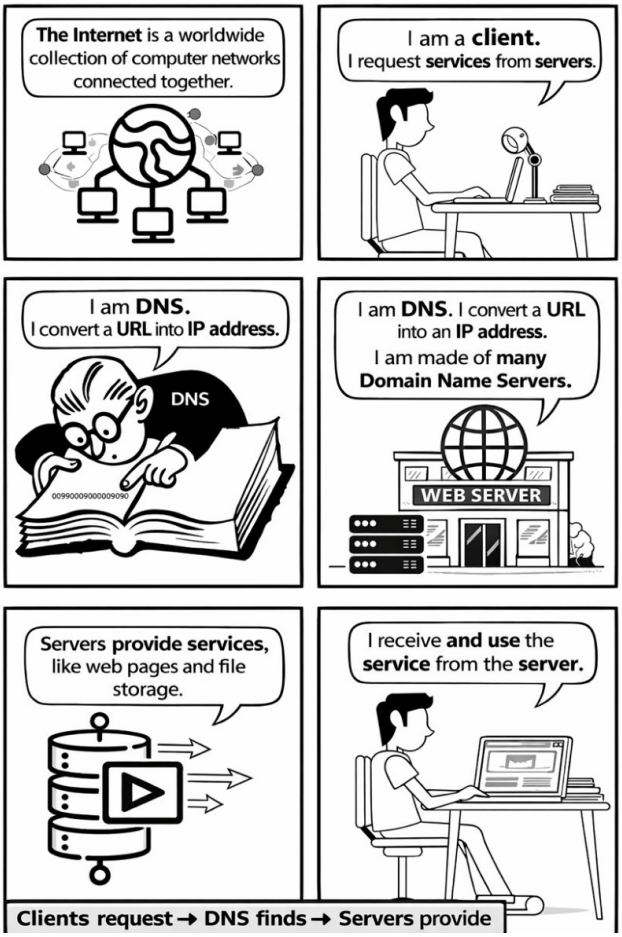
TECHNICAL ROLES · THE EXAM WORDS

NIC: provides network connectivity. Switch: connects devices on a LAN and forwards frames using MAC addresses. Router: connects networks and forwards IP packets between them. WAP: allows wireless devices to join a network. A single home router may combine several roles.

THEORY

The Internet, DNS and servers

Use the diagrams to identify connected networks, clients, servers and DNS.



HOW DNS HELPS YOU VISIT A WEBSITE

A browser uses the domain name in a web address to look up an IP address (unless already cached). DNS uses a distributed system of name servers; the browser can then contact the appropriate server.

CLIENTS AND SERVERS

A client requests a service, and a server responds. Web servers provide web content; file servers store and make files available. The Internet connects many networks; it is not one single server.

THEORY

Cloud services and network connections

Compare cloud storage, cloud software and cloud processing.

CLOUD STORAGE

Files reside on remote servers and can be accessed over a network, e.g. a shared school drive.

CLOUD SOFTWARE

Applications or functions are supplied remotely, e.g. a browser-based office service.

CLOUD PROCESSING

Remote servers carry out computational work, e.g. some online AI tools or rendering systems.

EXAM BALANCE · ADVANTAGES AND DISADVANTAGES

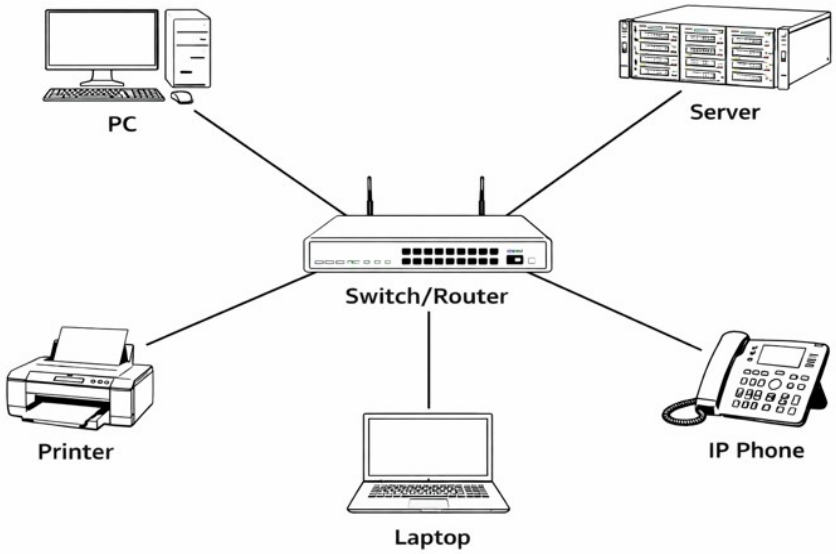
Potential advantages: access from different locations; shared work; less local storage; flexible capacity. Potential disadvantages: reliance on a connection and provider; ongoing cost; privacy/security considerations; reduced access during outages. Not every app or game sends all its work to a cloud server.

THEORY

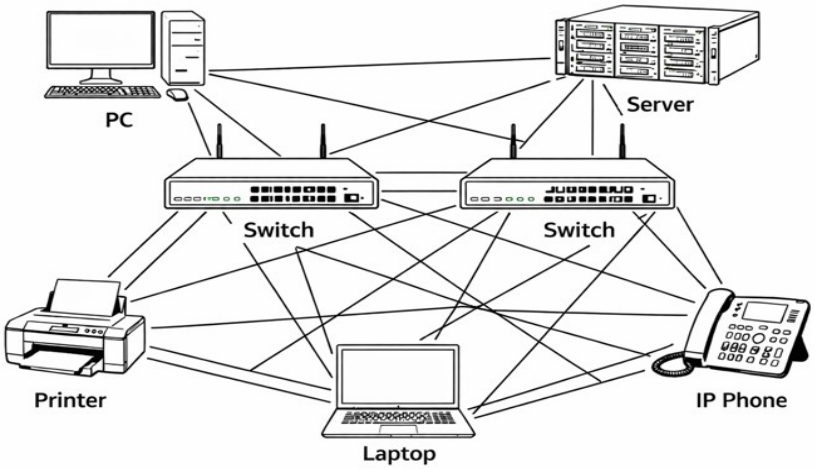
Star and mesh network topologies

Study the star and mesh diagrams, then describe their features.

STAR NETWORK / SCHOOL EXAMPLE



MESH NETWORK / MULTIPLE ROUTES



- Devices (or switches/routers) connected to multiple others
- Data can travel directly between nodes or wo alternative routes
- This makes them fault toralant

Star: devices link to a central network device; simple to manage but failure of that device can disconnect its attached devices. Mesh: nodes have multiple links, giving alternative paths, but requiring more connections and greater complexity. A mesh may be partial, not fully connected.

KEY TERMINOLOGY

Keywords for 1.3.1 · 1 of 2

The three yellow translucent pen strokes are taken from the shared J277/H446 lesson CSS.

Network	Connected devices sharing information, services or resources.
LAN	Local Area Network: a network covering a relatively small area.
WAN	Wide Area Network: links devices or networks across large distances.
Bandwidth	Maximum rate of data transfer, commonly measured in bits per second.
Latency	Time delay before data begins to reach its destination.
Client	Device or software that requests a service.
Server	Device or software providing a service to clients.
Peer-to-peer	Model where computers share resources directly with one another.

KEY TERMINOLOGY

Keywords for 1.3.1 · 2 of 2

The three yellow translucent pen strokes are taken from the shared J277/H446 lesson CSS.

NIC	Network interface component that connects a device to a network.
Switch	Connects LAN devices and forwards frames using MAC information.
Router	Forwards packets between networks using IP addresses.
WAP	Wireless access point, allowing wireless connections to a network.
DNS	Domain Name System: resolves domain names to IP addresses using name servers.
Cloud	Remote network-based storage, software or processing services.
Star topology	Network layout in which nodes connect to a central device.
Mesh topology	Layout where nodes have multiple connections and alternative paths.

WORKED EXAMPLES

Worked example 1 · The school network

Explain which network type and hardware a new school building might use.

SCENARIO

A school opens a second block on the same site. It wants classroom computers, a shared printer, Wi-Fi for tablets and Internet access.

STEP 1 · NAME THE TYPE

A LAN is appropriate because the school devices are on one local site.

STEP 2 · CONNECT THE COMPUTERS

NICs provide device connectivity. A switch joins wired devices within the LAN. A WAP allows tablets to connect wirelessly.

STEP 3 · CONNECT OTHER NETWORKS

A router connects the school LAN to an external network such as the Internet.

STEP 4 · LINK TO A REASON

Central devices help management, but the selected switch/WAP capacity and number of connected devices affect performance.

WORKED EXAMPLES

Worked example 2 · DNS lookup

Trace what happens when a student enters a web address into a browser.

STUDENT ENTERS A WEBSITE ADDRESS

The browser needs an IP address for the website domain name. The domain name is part of the URL.

1 · ASK

If a suitable answer is not already cached, a DNS lookup is made.

2 · LOOK UP

Distributed DNS name servers help resolve the domain name to an IP address.

3 · RETURN

The DNS lookup returns the address information to the client.

4 · CONNECT

The client can then use the returned address to contact the relevant web server.

WORKED EXAMPLES

Worked example 3 · Star versus mesh

Choose a topology for a school classroom and for a resilient network between sites.

SCHOOL CLASSROOM · STAR

A central switch connecting computers is usually practical: cabling and management are relatively straightforward. If that central switch fails, its connected computers lose their connection.

SITES NEEDING ALTERNATIVE ROUTES · MESH

Some networks benefit from multiple links between routers. Traffic can use another available route when one link fails. That improves resilience but adds hardware and complexity.

EXAM JUDGEMENT

There is no one best topology for every situation. Choose by cost, complexity, central management, reliability and need for alternative routes. A partial mesh can provide redundancy without connecting every device to every other device.

EXAM PRACTICE

Practice questions · set 1 of 5

Answer independently. Use the marks shown beside each question.

Q1 What is a computer network? **1 mark**

Q2 What does LAN stand for? **1 mark**

Q3 What does WAN stand for? **1 mark**

Q4 State two differences between a LAN and a WAN. **2 marks**

Q5 State two uses of a computer network. **2 marks**

EXAM PRACTICE

Practice questions · set 2 of 5

Answer independently. Use the marks shown beside each question.

Q6 Explain the difference between a client–server network and a peer-to-peer network. **3 marks**

Q7 Explain the role of a Network Interface Card (NIC). **3 marks**

Q8 Explain the role of a switch in a LAN. **3 marks**

Q9 Explain the difference between copper cable and fibre- optic cable. **3 marks**

Q10 Explain one disadvantage of wireless transmission. **3 marks**

EXAM PRACTICE

Practice questions · set 3 of 5

Answer independently. Use the marks shown beside each question.

Q11 Explain what DNS is used for. **3 marks**

Q12 Explain what is meant by the Cloud. **3 marks**

Q13 Explain one advantage of a star topology. **3 marks**

Q14 Explain one advantage of a mesh topology. **3 marks**

Q15 Describe the role of a router in a network. **4 marks**

EXAM PRACTICE

Practice questions · set 4 of 5

Answer independently. Use the marks shown beside each question.

Q16 Describe two advantages of a client– server network. **4 marks**

Q17 Describe two disadvantages of a peer-to- peer network. **4 marks**

Q18 Describe how DNS works. **4 marks**

Q19 Describe two cloud services. **4 marks**

EXAM PRACTICE

Practice questions · set 5 of 5

Answer independently. Use the marks shown beside each question.

Q20 Explain how the Internet works as a client–server network. **6 marks**

Q21 Compare star and mesh network topologies. **6 marks**

ANSWERS & MARK SCHEME

Indicative answers · set 1 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q1 · 1 MARK

What is a computer network?

Model answer: A network is two or more computers connected together to share data and resources.

Q2 · 1 MARK

What does LAN stand for?

Model answer: Local Area Network.

Q3 · 1 MARK

What does WAN stand for?

Model answer: Wide Area Network.

Q4 · 2 MARKS

State two differences between a LAN and a WAN.

Model answer: A LAN usually covers a smaller geographic area (e.g. one school site), whereas a WAN spans a much larger area and may link several LANs. Speed is not determined by network type alone.

Q5 · 2 MARKS

State two uses of a computer network.

Model answer: Networks allow file sharing and sharing hardware such as printers.

ANSWERS & MARK SCHEME

Indicative answers · set 2 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q6 · 3 MARKS

Explain the difference between a client–server network and a peer-to-peer network.

Model answer: In a client–server network, servers provide services and clients request them. In a peer-to-peer network, all computers have the same role.

Q7 · 3 MARKS

Explain the role of a Network Interface Card (NIC).

Model answer: A NIC allows a device to connect to a network and send and receive data, and it has a unique MAC address.

Q8 · 3 MARKS

Explain the role of a switch in a LAN.

Model answer: A switch connects devices in a LAN and sends data only to the intended device using MAC addresses.

Q9 · 3 MARKS

Explain the difference between copper cable and fibre- optic cable.

Model answer: Copper uses electrical signals and is slower, while fibre uses light signals and is much faster over long distances.

ANSWERS & MARK SCHEME

Indicative answers · set 3 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q10 · 3 MARKS

Explain one disadvantage of wireless transmission.

Model answer: Wireless transmission can suffer from interference and security issues.

Q11 · 3 MARKS

Explain what DNS is used for.

Model answer: DNS converts human-readable website names into IP addresses so computers can locate servers.

Q12 · 3 MARKS

Explain what is meant by the Cloud.

Model answer: The Cloud is the use of remote servers on the Internet to store data, run software, or process data.

Q13 · 3 MARKS

Explain one advantage of a star topology.

Model answer: Star networks are easy to manage because all devices connect to a central switch.

Q14 · 3 MARKS

Explain one advantage of a mesh topology.

Model answer: Mesh networks are fault tolerant because data can travel along multiple routes.

ANSWERS & MARK SCHEME

Indicative answers · set 4 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q15 · 4 MARKS

Describe the role of a router in a network.

Model answer: A router forwards data between different networks using IP addresses and connects a LAN to the Internet.

Q16 · 4 MARKS

Describe two advantages of a client– server network.

Model answer: Client–server networks allow central management and stronger security, with data backed up in one place.

Q17 · 4 MARKS

Describe two disadvantages of a peer-to- peer network.

Model answer: Peer-to-peer networks have weaker security and no central backup, and files may be unavailable if a computer is off.

Q18 · 4 MARKS

Describe how DNS works.

Model answer: A client sends a URL to a DNS server, which looks up the IP address and returns it so the client can contact the web server.

ANSWERS & MARK SCHEME

Indicative answers · set 5 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q19 · 4 MARKS

Describe two cloud services.

Model answer: Cloud storage stores files online, while cloud software allows applications to run in a web browser without installation.

Q20 · 6 MARKS

Explain how the Internet works as a client–server network.

Model answer: The Internet is a network of networks where clients request services from servers. DNS converts URLs into IP addresses, and servers provide services such as web pages and file storage.

Q21 · 6 MARKS

Compare star and mesh network topologies.

Model answer: Star networks are simple and easy to manage but rely on a central device. Mesh networks allow multiple routes, making them fault tolerant and reliable but more complex and expensive.

THEORY

Wired and wireless connections

Use the comparison diagram to connect each visual example to GCSE terminology.

WHAT STUDENTS NEED TO DECIDE

A wired connection uses a cable, while a wireless connection sends signals through the air. Choose by scenario: speed, interference, mobility, installation, reliability and security.

WIRED · ETHERNET

The cable is the fixed delivery route.

Often consistent for fixed equipment; cables limit movement and take time to install.

WIRELESS · WI-FI / BLUETOOTH

The connection works without a network cable to the device.

Useful for moving devices, but range, congestion and interference can affect performance.

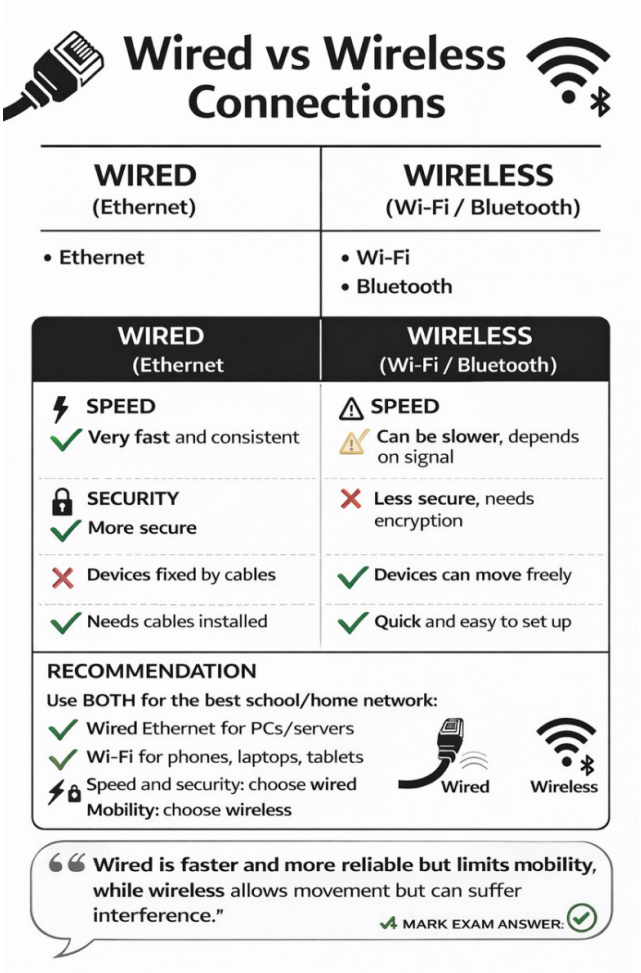
THE ANALOGY HAS LIMITS

A wired connection may be faster or more secure in some circumstances, but this is not guaranteed. Compare configuration, interference and encryption.

THEORY

Comparing wired and wireless networks

Use the categories to compare trade-offs rather than treating them as universal rankings.
universal rankings.



EXAM-ACCURATE COMPARISON

Wired: requires cables, usually offers stable high bandwidth to stationary devices. Wireless: allows movement and easier installation for mobile devices, but is affected by signal strength, interference and shared airtime.

RECOMMENDATION

For a school: Ethernet for desktop PCs and servers; Wi-Fi for tablets and laptops. Security depends on encryption, configuration and access control, not just whether a cable is present.

THEORY

Connections, signals and decisions

Select the connection for the task, then give a linked justification.

ETHERNET / COPPER

Electrical signals travel along copper cabling. Common for wired school LAN connections.

FIBRE OPTIC

Pulses of light travel through fibre. Useful for high-capacity links and long distances, but installation costs vary.

WI-FI

Uses radio for devices to connect to a wireless LAN through an access point.

BLUETOOTH

Short-range radio connection commonly used between nearby devices or peripherals.

THEORY

IP addresses and MAC addresses

Three distinct ideas: the IP version, what an address identifies, and who uses it.

IPv4

32 bits. Typically shown as four decimal values from 0 to 255 separated by dots. Example: 192.168.1.25. It is a logical network address used when delivering IP packets.

IPv6

128 bits. Typically expressed as eight groups of hexadecimal digits separated by colons; groups may be shortened. Example: 2001:db8::1. It addresses IPv4 address-space limitations.

MAC ADDRESS

A link-layer device/interface identifier used for local frame delivery, often represented using six groups of hexadecimal digits (48 bits), e.g. 00:1A:2B:3C:4D:5E. Many devices permit address randomisation or change.

NOT THE SAME AS DNS

An IP address identifies an interface/network destination for routing. A MAC address supports local link communication. DNS resolves domain names to address records; it does not turn an entire URL into one IP address.

THEORY

Encryption, standards and protocols

Data is vulnerable to interception in transit. Protocols and standards let systems communicate in understood

ENCRYPTION

Encryption transforms readable information into ciphertext using an algorithm and key. Without the correct key, an interceptor should not be able to understand the protected content. Encryption protects confidentiality; it does not alone prevent every attack.

STANDARDS

Agreed specifications let products from different manufacturers interoperate. For example, Wi-Fi and Ethernet standards allow compatible network equipment to exchange information.

COMMUNICATION PROTOCOLS

A protocol is a set of rules for communication, defining how messages are formatted, sent, received or handled. Different protocols perform different jobs: HTTP, HTTPS, TCP, IP, SMTP and others.

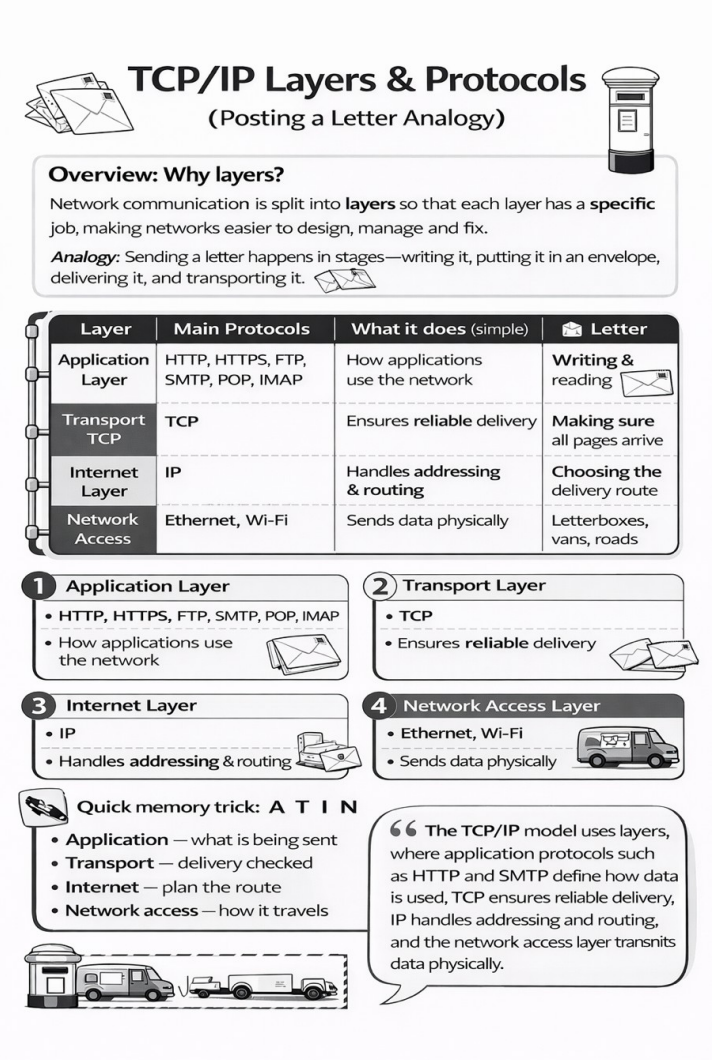
HTTPS IS MORE THAN JUST "HTTP + ENCRYPTION"

HTTPS protects HTTP traffic using TLS, also providing server authentication and integrity safeguards when correctly configured. A padlock does not prove the website itself is trustworthy.

THEORY

Protocols · postal delivery analogy

Use the postal analogy to remember each purpose, then explain it in technical terms.



WHAT THE ANALOGY EXPLAINS

Application: writing/reading the letter (the service requested). Transport: checking the pages arrive in order. Internet: selecting destination and route. Network access: actual movement across the link. OCR note: the names and functions of individual TCP/IP layers are not required for memorisation in J277, but the benefit of using layers is required.

THEORY

Which protocol does what?

Use the postal analogy,

HTTP	Request and transfer web resources; not encrypted by default.
HTTPS	HTTP over TLS; protects web communications in transit.
FTP	Transfer files between networked systems.
SMTP	Send and relay email messages.
POP3	Retrieve/download messages from a mail server; copies can remain depending on settings.
IMAP	Access and synchronise email kept on the server across devices.
TCP	Reliable ordered byte-stream delivery with error detection and retransmission.
IP	Logical packet addressing and forwarding between networks.

KEY TERMINOLOGY

Key words for 1.3.2 · 1 of 2

Highlights use the three exact translucent yellow SVG strokes from the shared J277/H446 lesson CSS.

Ethernet	Wired local-network technology using defined standards.
Wi-Fi	Wireless local networking using radio signals.
Bluetooth	Short-range wireless communication commonly between nearby devices.
Fibre optic	Cable transmitting light pulses to carry digital data.
Interference	Unwanted signals that disrupt or reduce transmission quality.
Encryption	Transforms plaintext into ciphertext using cryptographic methods/keys.
Ciphertext	Encrypted information that should be unreadable without the required key.
Standard	Agreed specification allowing compatible systems to work together.
Protocol	Rules that systems follow when communicating.

KEY TERMINOLOGY

Key words for 1.3.2 · 2 of 2

Highlights use the three exact translucent yellow SVG strokes from the shared J277/H446 lesson CSS.

IPv4	32-bit Internet Protocol address format, commonly four decimal octets.
IPv6	128-bit Internet Protocol address format, commonly hexadecimal groups.
MAC address	Link-layer identifier used in the delivery of local network frames.
HTTP / HTTPS	Web transfer protocol / HTTP protected using TLS.
SMTP	Protocol for sending or relaying email.
POP3	Protocol commonly used for downloading email from a server.
IMAP	Protocol for synchronising email held on a server.
TCP	Protocol offering reliable ordered delivery for applications.
IP	Protocol providing logical packet addressing and routing between networks.

WORKED EXAMPLES

Example 1 · Choosing connections

A school needs reliable fixed PCs and roaming tablet devices.

SCENARIO

A computer room has 28 desktop PCs. Pupils also use tablets across classrooms. Recommend how each device type should connect.

STEP 1 · FIXED COMPUTERS

Choose wired Ethernet: no need to move the PCs and a cable can give stable, high-capacity links.

STEP 2 · MOBILE TABLETS

Choose Wi-Fi: pupils can move between suitable access points without network cables.

STEP 3 · THINK ABOUT REAL LIMITATIONS

Wi-Fi performance depends on signal strength, interference and user numbers. Ethernet needs installation and switch ports.

STEP 4 · EXPLAIN SECURITY

Apply suitable network controls and encryption. Never assume wired is automatically secure, or Wi-Fi automatically unsafe.

WORKED EXAMPLES

Example 2 · Addressing and delivery

Do not confuse a MAC address, an IP address and DNS name resolution.

STEP 1 · IDENTIFY IP VERSION

192.168.5.19 is an IPv4 address: four decimal octets, each between 0 and 255.

STEP 2 · RECOGNISE AN IPv6 FORMAT

2001:db8::5 is a valid shortened IPv6 notation using hexadecimal and colons.

STEP 3 · CONNECT THE MAC ADDRESS

A switch uses the local link-layer destination MAC address to forward Ethernet frames on a LAN.

STEP 4 · CONNECT IP ROUTING

A router examines logical IP destinations to forward packets between networks.

STEP 5 · CONNECT DNS

DNS can resolve a domain name to an IP address; it does not translate an entire webpage URL.

WORKED EXAMPLES

Example 3 · Posting a secure letter

Use the analogy to explain separate jobs performed by network protocols.

SCENARIO

A student sends an email containing a homework file, then opens a website to check the deadline.

SMTP · PUTTING THE LETTER IN THE POST

SMTP sends and relays messages. It is not the protocol normally used to read a synced inbox.

IMAP · CHECKING THE POSTBOX

IMAP lets the student access and keep mail synchronised on several devices.

TCP/IP · GETTING THE LETTER THERE

TCP can support reliable ordered transport; IP carries addressing and routing information between networks.

HTTPS · THE SECURE ENVELOPE

HTTPS uses TLS to protect information exchanged with the website. Never claim it guarantees the content or site is harmless.

EXAM PRACTICE

Independent questions · 1 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

Q1 **1 mark**

What is encryption?

Q2 **1 mark**

Name one communication protocol.

Q3 **1 mark**

What does HTTPS provide that HTTP does not?

Q4 **2 marks**

State two reasons encryption is needed on Wi-Fi networks.

Q5 **2 marks**

State two benefits of network standards.

EXAM PRACTICE

Independent questions · 2 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

Q6 **3 marks**

Explain why data sent over the Internet should be encrypted.

Q7 **3 marks**

Explain the purpose of communication protocols.

Q8 **3 marks**

Explain the difference between HTTP and HTTPS.

Q9 **3 marks**

Explain the difference between SMTP and POP.

Q10 **3 marks**

Explain the difference between POP and IMAP.

EXAM PRACTICE

Independent questions · 3 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

Q11 **4 marks**

Describe the role of TCP in data transmission.

Q12 **4 marks**

Describe the role of IP in network communication.

Q13 **4 marks**

Describe two differences between wired and wireless networks.

Q14 **6 marks**

Explain how protocols and standards allow data to be transmitted successfully across networks.

EXAM PRACTICE

Independent questions · 4 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

E1 · EXTRA OCR COVERAGE — **2 marks**

State the number of bits in an IPv4 address and describe the usual written format.

E2 · EXTRA OCR COVERAGE — **2 marks**

State the number of bits in an IPv6 address and describe how its value is written.

E3 · EXTRA OCR COVERAGE — **3 marks**

Explain why a switch uses a MAC address but a router uses an IP address when forwarding traffic.

E4 · EXTRA OCR COVERAGE — **3 marks**

A school needs fixed desktop computers in a lab and mobile tablets in classrooms. Recommend suitable connections and justify each.

ANSWERS & MARK SCHEME

Indicative answers · 1 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

Q1 — 1 mark

Encryption is the process of scrambling data so that only authorised users can read it.

Q2 — 1 mark

HTTP / HTTPS / FTP / SMTP / TCP / IP.

Q3 — 1 mark

Encryption / security.

Q4 — 2 marks

Wi-Fi data travels through the air and can be intercepted, so encryption prevents unauthorised access.

Q5 — 2 marks

Standards allow devices from different manufacturers to work together and ensure reliable communication.

ANSWERS & MARK SCHEME

Indicative answers · 2 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

Q6 3 marks

Data passes through many routers and could be intercepted, so encryption prevents unauthorised users from reading it.

Q7 3 marks

Protocols are sets of rules that control how data is transmitted so devices can communicate correctly.

Q8 3 marks

HTTP is used to view web pages, while HTTPS encrypts the data to provide secure communication.

Q9 3 marks

POP3 downloads messages from a server. Whether they remain on the server depends on settings.

Q10 3 marks

POP downloads emails to one device, while IMAP synchronises emails across multiple devices.
Exam accuracy: State that IMAP normally synchronises mail stored on the server, while POP3 commonly retrieves/downloads it.

ANSWERS & MARK SCHEME

Indicative answers · 3 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

Q11 4 marks

TCP ensures data is delivered reliably and in the correct order by checking that packets arrive successfully.

Q12 4 marks

IP is responsible for addressing and routing data so it reaches the correct destination.

Q13 4 marks

Wired networks use cables and are more reliable, while wireless networks use radio waves and are more flexible but can suffer interference.
Exam accuracy: Avoid universal performance rankings; wired is generally stable and requires cables, while wireless permits mobility but can face interference.

Q14 6 marks

Standards ensure devices from different manufacturers can communicate. Protocols such as HTTP, TCP, and IP define how data is requested, delivered reliably, and routed to the correct destination, making communication efficient and reliable.

E1 · EXTRA 2 marks

IPv4 uses 32 bits (1); the usual format is four decimal numbers separated by dots, with each value from 0 to 255 (1).

ANSWERS & MARK SCHEME

Indicative answers · 4 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

E2 · EXTRA — **2 marks**

IPv6 uses 128 bits (1); it is generally written as eight groups of hexadecimal digits separated by colons, with shortening permitted (1).

E3 · EXTRA — **3 marks**

A switch uses link-layer MAC addresses to deliver frames locally (1). A router uses logical IP addresses to forward packets between networks (1). Their jobs differ by network scope/layer (1).

E4 · EXTRA — **3 marks**

Use wired Ethernet for fixed desktops for a stable, high-capacity connection (1); use Wi-Fi for mobile tablets to allow movement (1). Explain that encryption, access point capacity and signal strength must also be considered (1).